

SQL

Theory Notes

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1. TCL – Transaction Control Language

- **Definition:** Commands that manage transactions in a database.
- **Key commands:**
 - `START TRANSACTION` → begins a transaction.
 - `COMMIT` → saves changes permanently.
 - `ROLLBACK` → undoes changes to the last committed state.
 - `SAVEPOINT` → sets a point within a transaction to roll back to.
 - `ROLLBACK TO SAVEPOINT` → undoes changes only up to the savepoint.
- **Use case:** Ensures data consistency and allows safe recovery from errors.

2. DCL – Data Control Language

- **Definition:** Commands that control access to data in the database.
- **Key commands:**
 - `GRANT` → gives privileges to a user.
 - `REVOKE` → removes privileges from a user.
- **Privileges:** `SELECT`, `INSERT`, `UPDATE`, `DELETE`, `ALTER`, `ALL`.
- **Use case:** Security and access control in multi-user environments.

Coding Round Questions

TCL

1. Create a table `accounts` with fields `id` and `balance`. Insert 5 records.
2. Start a transaction and update all balances to 1000. Use `ROLLBACK` to undo changes.
3. Start another transaction, update balance of `id=1` to 50000, set a savepoint, then update balance of `id=2` to 30000. Roll back to the savepoint and commit.

DCL

4. Create two users `user1` and `user2` with passwords.
5. Grant `SELECT` privilege on a table `departments` to `user2`.
6. Grant `ALL` privileges on the database to `user2`.
7. Revoke `SELECT` privilege from `user2`.
8. Drop `user2` from the database.

```
-- tcl => TRANSACTION controll language

CREATE TABLE accounts(
    ID INT,
    balance DOUBLE
);

INSERT INTO accounts VALUES(1,40000),(2,50000),(3,60000),(4,10000);

SELECT * FROM accounts;

DELETE FROM accounts;

START TRANSACTION;

UPDATE accounts SET balance = 1000;

COMMIT; --SAVES PERMANENTLY

ROLLBACK; -- ROLL BACK TO PREVIOUS STATE

START TRANSACTION;
UPDATE accounts SET balance = 50000 WHERE id=1;

SAVEPOINT sp1;

UPDATE accounts SET balance = 30000 WHERE id=2;

ROLLBACK TO sp1;

COMMIT;

-- dcl -> data control language--

CREATE user 'pirithika'@'localhost' identified BY '1234';

CREATE user 'maha'@'localhost' identified BY '1234';

GRANT SELECT ON mysqlpirithika.departments TO 'maha'@'localhost';

GRANT ALTER ON mysqlpirithika.departments TO 'maha'@'localhost';

GRANT UPDATE ON mysqlpirithika.departments TO 'maha'@'localhost';

GRANT DELETE ON mysqlpirithika.departments TO 'maha'@'localhost';
```

```
SHOW GRANTS FOR 'maha'@'localhost';
```

```
GRANT ALL ON mysqlpirithika.* TO 'maha'@'localhost';
```

```
REVOKE SELECT ON mysqlpirithika.* FROM 'maha'@'localhost';
```

```
REVOKE ALL privileges ON mysqlpirithika.* FROM 'maha'@'localhost';
```

```
DROP user 'maha'@'localhost';
```